

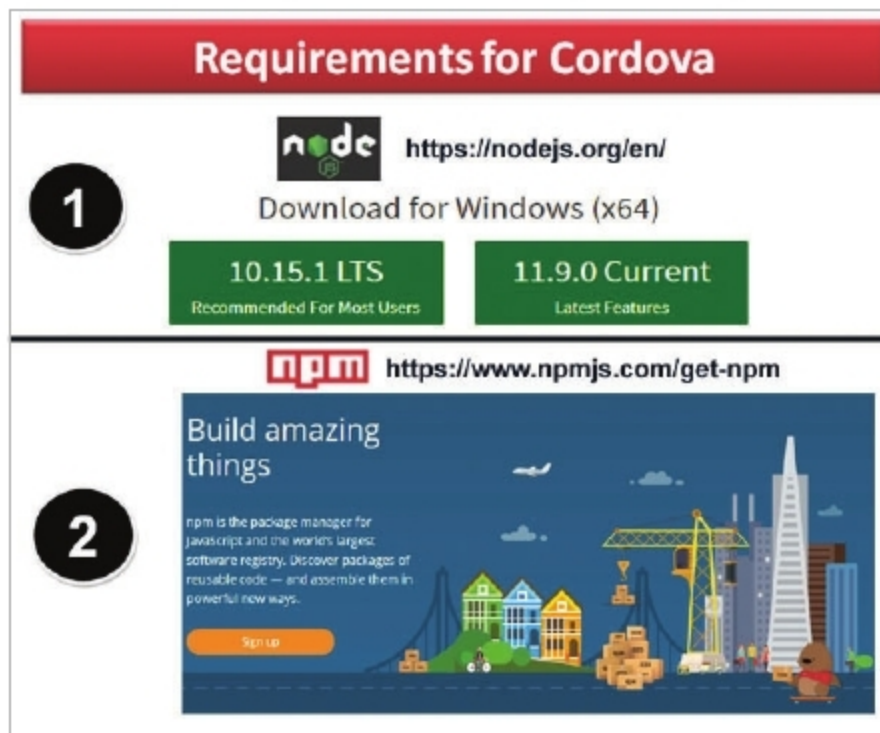


Figure 3: Requirements for Apache Cordova



Figure 4: Searching and adding plugins for multiple platforms

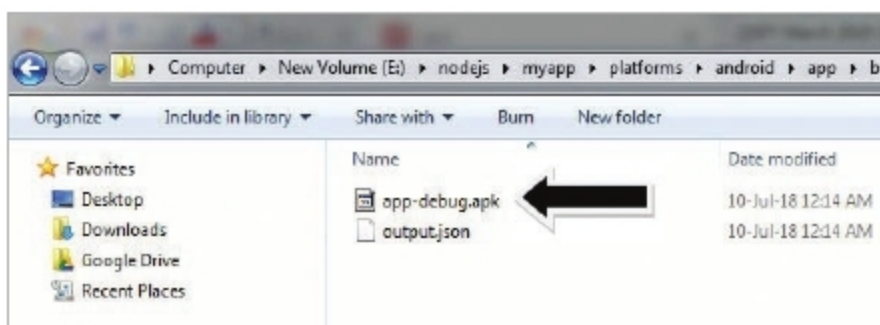


Figure 5: APK file generated by Apache Cordova

and this obviously undermines profitability.

Installation and development of cross-platform mobile apps using Apache Cordova

Apache Cordova (<https://cordova.apache.org/>) is a popular and effective development platform for cross-platform as well as hybrid app development. With Cordova, mobile apps can be developed using HTML, JS and CSS, and the same code can be used for multiple mobile platforms.

Cordova can be programmed on the command line using Node.js (<http://nodejs.org/>). It can be downloaded using NPM

(<https://npmjs.org/package/cordova>). Apache Cordova can be installed for multiple platforms including Android, Windows, iOS or OSX. A number of plugins and libraries are available to integrate additional features in the smartphone app.

Using the command prompt, the installation of Apache Cordova can be done as follows.

For development on Windows, use the following command:

```
Windows Drive:>npm install -g cordova
```

For Linux/Mac OS X, use the following command:

```
$ sudo npm install -g cordova
```

Creating a new project in Apache Cordova after installing Node.js: Use the following code to begin creating a new project:

```
E:\nodejs> cordova create mymobileapp xxxxxx.yyyyyy.zzzzzz
MyApp
```

```
E:\nodejs> cd mymobileapp
```

To add the mobile platform for compatibility, use the following command:

```
E:\nodejs> cordova platform add android
E:\nodejs> cordova platform add ios
```

To view the installed platforms, use the command given below:

```
E:\nodejs> cordova platform ls
```

For the output, use the following code:

```
Installed platforms:
android 7.0.0
ios 4.5.4
```

```
Available platforms:
browser ~5.0.1
OSX ~4.0.1
```

```
windows ~5.0.0
```

```
www ~3.12.0
```

To add plugins and modules for integration of the app, use the following commands:

```
E:\nodejs> cordova plugin search camera
E:\nodejs> cordova plugin add cordova-plugin-camera
```